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# A MATHEMATICAL FRAMEWORK FOR CONTACT DETECTION BETWEEN QUADRIC AND SUPERQUADRIC SURFACES

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**Abstract.** *Distance calculation between surfaces plays an important role in computational mechanics, namely, in the study of constrained multibody systems where contact forces take part. In this paper, a general contact detection methodology for non-conformal surfaces, described as quadric and superquadric surfaces, is presented. The mathematical framework relies on algebraic and differential geometry, vector calculus, and on the $C^2$ continuous implicit and parametric representations of the surfaces. The proposed methodology establishes a set of collinear and orthogonal contraints between vectors defining the contacting surfaces that, allied to loci contraints, formulate the contact problem. This set of non-linear equations is solved numerically with the Newton-Raphson method for which the Jacobian matrices are calculated analytically. The outputs consists on the coordinates of the pair of points with common normal vectors and, consequently, the minimum distance between both surfaces. Reactive forces appear whenever contact occurs and are inserted within the framework of multibody dynamics. The normal contact forces are evaluated, here, with a continuous contact force model, which is based on the Hertzian elastic contact and on the Hunt and Crossley dissipation model. For mechanical systems presenting more complex shapes, that are more accurately modeled with freeform surfaces, a fitting algorithm is exploited to approximate a superquadric to a set of points that belong to the generalized surface. Basically, the fitting algorithm consists on the optimization problem of minimizing the distances between the points within the neighbourhood of the contact site and the approximate surface. Several mathematical and computational aspects, regarding the contact detection methodology, are described.*

# 1 INTRODUCTION

Contact is an omnipresent phenomenon in any mechanical system. Physically, it can be defined as the tendency of two bodies on sharing a common locus in space in which reactive forces are generated to oppose body intersection and energy is dissipated, usually, in the form of heat.

To accurately design and simulate a mechanism, that is either interacting with the surrounding environment or devising relative joint motion among the articulated bodies that compose it, contact forces must be utterly considered [1]. Within multibody dynamics, contact analysis is a computational methodology used to describe the behaviour of a constrained mechanical system induced by reactive forces that are produced during body collision. Contact analysis focuses on the resolution of three fundamental issues: (i) minimum distance calculation between surfaces; (ii) contact detection; and (iii) establishment of a force model that depends on material properties and (pseudo-)penetration depth. Such analysis requires a representative geometric description of the bodies boundaries.

Given a non-conformal (or convex) surface representation of the colliding bodies, together with their position and spatial orientation, the resolution of the minimum distance calculation problem is purely geometric. The unknown variables of such problem are the spatial coordinates of the candidate contacting pair of points. Usually, contact between two surfaces is established by the resolution of a set of non-linear equations that expresses collinear and orthogonal constraints between vectors defining the contacting surfaces, namely, the normal, tangent, binormal and distance vectors [1-3]. Another type of equation that is important for the geometric accuracy of contact analysis is hereafter called as the locus condition or isosurface condition (isoline condition, in two-dimensional space). The solution of this type of equations are the points that satisfy a geometric property. As an example, the locus condition of a sphere is the location of all the points equally distanced from a point, i.e., the center point.

The intersection of rigid bodies geometries, although physically impossible, can be considered as a motif for the mathematical concept of contact: two surfaces (or lines) are in contact when their intersection is not a null set of points. Contact detection consists on determining if the referred bodies are in physical contact, thus, three contact status are possible: (i) no contact; (ii) external contact or contact at a sigle point; and (iii) contact with pseudo-penetration. From a mechanical point of view, whenever contact occurs it is said that the rigid bodies overlap or present pseudo-penetration. Here, contact detection relies on evaluating the minimum distance between surfaces, i.e., contact is detected when the minimum distance is lesser or equal to zero and positive when surfaces are apart. By convention, negative distances imply surface overlap.

The magnitude of the contact reaction forces, which depends on the minimum distance, can be calculated only when pseudo-penetration occurs. Within the equations of motion, contact forces are seen as external forces that act upon interacting or interlinked bodies. Different formalisms can be used to describe the continuous contact forces, including penalty and linear complementary contact formulations. A common model for the continuous contact forces was proposed by Lankarani and Nikravesh, which is based on the Hertzian non-linear elastic contact [4]. Hunt and Crossley proposed a more general and continuous contact force model that also accounts for energy disspation [5]. Although other formalisms, such as the linear complementary problem, can be used to evaluate the contact forces, taking advantage of the proposed geometric description of contact presented here, such methodologies are not considered in this work.

For non-conformal contact analysis, the surface associated to the rigid body must be described with a mathematically well-defined and $C^2$ continuous geometric representation, so that analytical expressions of the distance vector and the normal, tangential and binormal surface vectors can be deduced. The surface representation can be either implicit or explicit (parametric). This being, the mathematical framework presented hereafter relies on vector calculus, algebraic and differential geometry. One of the main requirements to model the geometry of a three-dimensional object, in this case the outer surface of a rigid body, is the usage of a mathematical description that provides high geometric representativity. Quadric or superquadric surfaces are geometric entities that provide such a description for a variety of shapes, both natural and manmade objects.

For mechanical systems presenting freeform shapes, superquadrics surfaces can be fitted to a set of points that belong to the generalized surface by solving the following optimization problem: minimizing the distances between the points within the neighbourhood of the contact site and the approximate surface. This strategy is suitable to handle contact problems in which the contacting surfaces either have geometries close to a superquadratic surface or the contact points do not depart from a small neigborhood of the surface.

Contact analysis has many important applications in other areas of applied sciences such as, molecule simulation in computational physics [6], modeling discontinuous mechanical systems (discrete element method) in geomechanics [7], humanoid design in biomechanics [8], virtual reality simualtion and computer animation. From such a variety of applications, several methods have been developed for distance computation and contact detection based on quadric and superquadric surfaces.

Based on an interior point algorithm, Chakraborty et al. [9] formulated the distance computation for convex bodies as an optimization problem to minimize the euclidean distance subjected to the condition that candidate points lie on the surfaces. Such method presents global convergence properties that are robust even in the absence of any initial information about the closest points. The method is also quite accurate since the determined points belong to the surface. Its main drawback is a lesser computational efficiency compared to other methods [7,8].

For discrete element modeling, Lin and Ng [7] applied contact detection algorithms for ellipsoids based on the common normal vector concept, which states that two points are the candidate contact pair of points if the normal directions at these points are parallel to the intersecting line. A set of non-linear equations obtained from equaling both normal vectors and from equaling the normal vector to the distance vector, is solved numerically. Two additional conditions of the points lying on the ellipsoids are also considered. This method was tested only for ellipsoids and does not regard other vector relationships that are fundamental for numerical robustness and accuracy.

For biomechanical applications, Kwak et al. [8] developed multibody models of diarthrodial joints, such as the knee. The models include articular contact of freeform surfaces with a parametric representation. Contact detection is also founded on the common normal vector concept. Here, it is pointed out that the computational efficiency of their contact model is enhanced by the use of analytical Jacobians.

Wang et al. [10] presented an algebraic condition for real-time continuous collision detection for ellipsoids moving under rigid body transformations. The algebraic condition is a quartic polynomial equation, also named as separation condition or characteristic equation, that relates the geometric parameters of shape, spatial orientation, and position of two ellipsoids. Depending on the sign of all four roots it is possible to determine the contact status. The resolution of the characteristic equation is strightfoward, leading to a simple and yet efficient algorithm for contact detection of ellipsoidal bodies that computes the exact time interval of the collision [11].

In this paper, a general non-conformal contact detection methodology for rigid bodies, described by quadric and superquadric surfaces, is presented. The proposed contact detection methodology establishes a set of geometric conditions that, once fulfilled, render the pair of points of possible contact. The resulting methodology consists in a set of non-linear equations that is solved numerically with the Newton-Raphson method for which the Jacobian matrices are calculated analytically, without resorting to optimization algorithms. This way, contact detection comes naturally from the minimum distance calculation. The theoretical formulations of the geometric description and the computational implementation aspects of contact detection are described in the following sections.

## 2 CONTACT DETECTION METHODOLOGY AND MULTIBOY DYNAMICS

Within the multibody dynamics formulation, the second-order ordinary differential equations that describe the motion of a constrained mechanical system are given by:

$$\begin{cases} \mathbf{M}\ddot{\mathbf{q}} + \mathbf{\Phi}_{\mathbf{q}}^{T}\boldsymbol{\lambda} = \mathbf{g} \\ \mathbf{\Phi}_{\mathbf{q}}\ddot{\mathbf{q}} = \boldsymbol{\gamma} \end{cases} \tag{1}$$

where $\mathbf{M}$ is the global mass matrix, $\ddot{\mathbf{q}}$ is the acceleration vector, $\mathbf{\Phi}_{\mathbf{q}}$ is the Jacobian matrix of the kinematic constraints, $\boldsymbol{\lambda}$ is the vector of Lagrangian multipliers, $\boldsymbol{\gamma}$ is the right-hand side vector of the acceleration equation and $\mathbf{g}$ is the external force vector. Contact forces are, therefore, included in the $\mathbf{g}$ vector that contains all the external forces applied to the system. The relative position and orientation of the bodies are obtained by integrating in time the equations of motion given by Eq. 1. Spatial data, such as translational coordinates and euler angles, varies with time and are contained within vector $\mathbf{q}$, the vector of generalized coordinates.

Contact mechanics relies on the geometric constraint that expresses the common normal principle (although Lin and Ng [7] have referred it as a ‘concept’ it should be understood as a geometric principle). Restriction to body movement by such a constraint is imposed by the application of a pair of forces that appear when surfaces overlap, and not by the conditioning of a DOF within the mechanical system.

For every time step, contact detection is evaluated by determining the minimum Euclidean distance between the surfaces or, to test if the bodies are apart, by evaluating proximity queries [9]. When two moving surfaces overlap, a pair of opposite contact normal forces appear, each one applied on the contact points of the intervining bodies and with the direction of the minimum distance vector. The reaction force magnitude is proportional to the pseudo-penetration. Although the contact detection methodology regards bodies as being rigid, the pseudo-penetration is an estimate of the deformation that results from the overlap of the surfaces. This way, not only is it possible to predict the contact forces but also to estimate the contact areas and contact stresses.

In Fig. 1, a generalized multibody model [1] of a contact pair is presented. Each rigid body is defined with at least three points and can contain more than one contact surface. The coordinate systems employed to describe the motion and configuration of a contact pair are: the global coordinate system XYZ; the coordinate systems $\xi_k\eta_k\zeta_k$ of rigid bodies $k \in \{\alpha,\beta\}$; and the local coordinate systems $x_m y_m z_m$ of surfaces $m \in \{i,j\}$. Position vectors and transformation matrices, obtained from the multibody dynamics calculations, are represented, respectively, by $\mathbf{r}$ and $\mathbf{A}$ where the subscripts indicate the corresponding coordinate systems, e.g., $\mathbf{A}_{O\beta}$ represents the coordinate transformation from the local coordinate system of body $\beta$ to the global coordinate system. Position vectors in the local reference system of the contacting surface are represented by $\mathbf{s}$. Each body has a fixed coordinate system that must not be confused with the local coordinate systems of the surfaces that compose it. Note that, contact calculation is performed using the outward surface normals and that all vectorial relationships, that define the geometric constraints, are referenced to the global coordinate system.

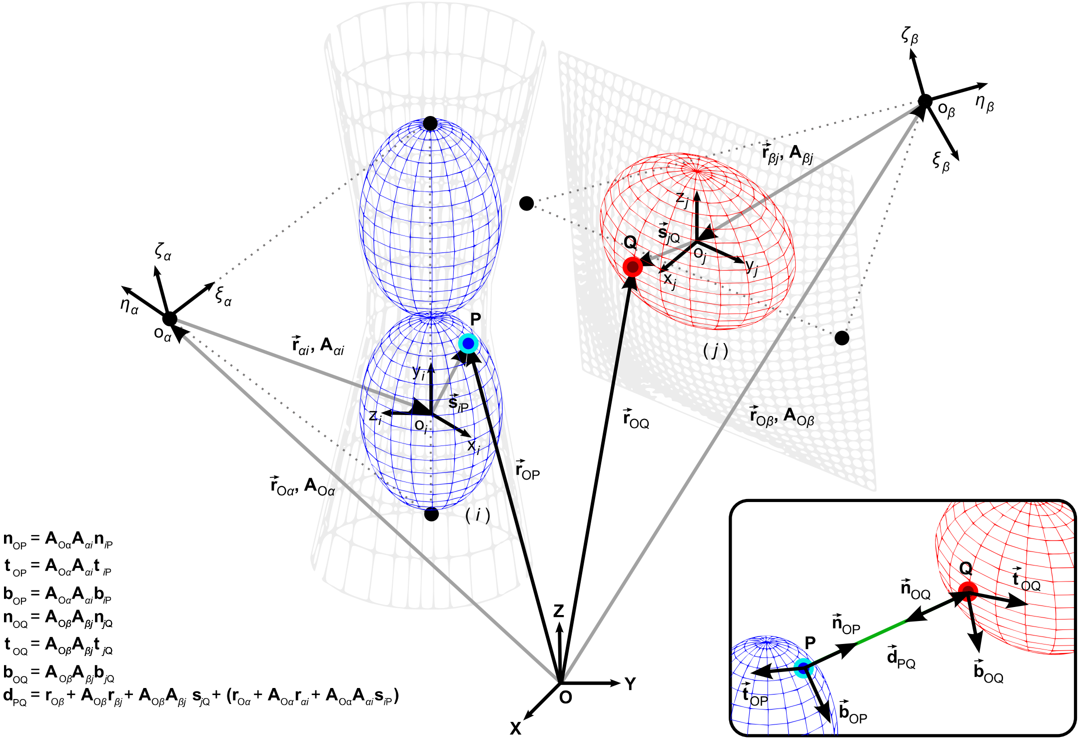


Figure 1: Multibody model of a contact pair and vector entities involved in contact analysis.

## 3 MATHEMATICAL FORMULATION FOR DISTANCE CALCULATION AND CONTACT DETECTION

Contact detection between two rigid and convex surfaces is a geometric problem that has an unique solution. Accordingly, consider two non-deformable bodies $\alpha$ and $\beta$ whose global positions and orientations are given. Rigidly coupled to each body are non-conformal $C^2$ continuous surfaces $i$ and $j$ that bear implicit and/or parametric representations. The common normal principle states that two points are a contact pair, if and only if, the normal vectors at these points are collinear relatively to each other and collinear with the minimum distance vector that connects the two points. From this enunciation, one can infer that contact formulation is intimately related to the surface representation. Implict and parametric surface representations are written, in a general form, as:

$$
\begin{aligned}
\mathcal{F}_i\left(x_i, y_i, z_i\right) &= 0, \\
\mathcal{F}_j\left(x_j, y_j, z_j\right) &= 0,
\end{aligned}
\tag{2}
$$

and

$$
\begin{aligned}
\mathbf{p}_i &= \mathbf{p}_i\left(s, u\right), \\
\mathbf{p}_j &= \mathbf{p}_j\left(w, v\right),
\end{aligned}
\tag{3}
$$

where $\mathcal{F}_i$ and $\mathcal{F}_j$ are implicit scalar functions that define the locus of points belonging to surfaces $i$ and $j$, respectively; $\mathbf{p}_i$ and $\mathbf{p}_j$ are the parametric position vectors of the referred surfaces; $(x,y,z)$ the cartesian coordinates of a point belonging to the surface; and $(s,u)$ and $(w,v)$ parametric coordinates. The distance vector that connects the two contact points candidates located on the surfaces, $\mathbf{d}_{\mathrm{PQ}}$, is defined as the difference of the global vector positions of points P and Q, given by $\mathbf{r}_{\mathrm{OP}}$ and $\mathbf{r}_{\mathrm{OQ}}$. Thus, one has,

$$\mathbf{d}_{\mathrm{PQ}} = \mathbf{r}_{\mathrm{OQ}} - \mathbf{r}_{\mathrm{OP}}, \tag{4}$$

where $\mathbf{r}_{\mathrm{OP}}$ and $\mathbf{r}_{\mathrm{OQ}}$ are unknown quantities. The normal vector at each point is derived by partial differentiation of the surface equation in order to the spatial coordinates. If the implicit surface representation is considered (Eq. 2), the normal vector is the surface gradient. A plane that contains the tangent vectors can then be deduced since a plane can be defined by a point and a normal vector. In the case of parametric surfaces (Eq. 3), differentiation of the surface expression gives the tangential vectors along each spatial parameter. Therefore, surface normals are calculated by the cross product of the tangent vectors.

Contact detection is formulated by a set of *geometric constraints*, including the aforementioned common normal principle, which is written as a vector $\mathbf{\Phi}^G = \mathbf{\Phi}^G(\mathbf{q}^G)$ that depends on the vector of the unknown points,

$$\mathbf{q}^G = \begin{bmatrix} \mathbf{r}_{\mathrm{OP}} \\ \mathbf{r}_{\mathrm{OQ}} \end{bmatrix}. \tag{5}$$

For convinience of the methodology, vector $\mathbf{q}^G$ can also be expressed in local coordinates of the intervining surfaces, as it will be described in the next section. In general, three types of non-linear algebraic equations define a geometric constraint:

- orthogonal constraint (relative constraint between two perpendicular vectors);
- collinear contraint (relative constraint between two parallel vectors);
- locus or isosurface constraint (a point in space that lies on a contact surface).

It should be noticed that the latter type of constraint is only introduced for contact detection regarding implicit surfaces.

As it was mentioned before, contact between surfaces is enunciated as the problematic of encountering the location of the pair of points in which the distance vector, $\mathbf{d}_{\mathrm{PQ}}$, is aligned to the surface normals $\mathbf{n}_{i\mathrm{P}}$ and $\mathbf{n}_{j\mathrm{Q}}$, and has a minimum value. This can be written as two cross product equations relating vectors $\mathbf{d}_{\mathrm{PQ}}$, $\mathbf{n}_{i\mathrm{P}}$, and $\mathbf{n}_{j\mathrm{Q}}$. Equivalently, the collinearity condition can be expressed as an orthogonal constraint involving vector $\mathbf{d}_{\mathrm{PQ}}$, surface tangent vectors $\mathbf{t}_{i\mathrm{P}}$ and $\mathbf{t}_{j\mathrm{Q}}$, and binormal vectors $\mathbf{b}_{i\mathrm{P}}$ and $\mathbf{b}_{j\mathrm{Q}}$. Eq. 6 presents the equivalency of both constraints,

$$\begin{aligned} \mathbf{d}_{\mathrm{PQ}} \times \mathbf{n}_{i\mathrm{P}} = \mathbf{0} \;&\Leftrightarrow\; \begin{cases} \mathbf{d}_{\mathrm{PQ}} \cdot \mathbf{t}_{i\mathrm{P}} = 0 \\ \mathbf{d}_{\mathrm{PQ}} \cdot \mathbf{b}_{i\mathrm{P}} = 0 \end{cases} \\ \mathbf{d}_{\mathrm{PQ}} \times \mathbf{n}_{j\mathrm{Q}} = \mathbf{0} \;&\Leftrightarrow\; \begin{cases} \mathbf{d}_{\mathrm{PQ}} \cdot \mathbf{t}_{j\mathrm{Q}} = 0 \\ \mathbf{d}_{\mathrm{PQ}} \cdot \mathbf{b}_{j\mathrm{Q}} = 0 \end{cases} \end{aligned}. \tag{6}$$

In 3-D space, whether one considers the collinear or orthogonal constraint definition, Eq. 6 always renders a system of four non-linear independent equations. According to the orthogonal constraint, Eq. 6 can be solved for the 4 unknown surface parameters $(s,u,w,v)$ in the case of parametric surfaces, or when the surface representation follows an implicit definition, 2 additional locus contraints given by Eq. 2, one for each rigid body, must be added to the set of orthogonal constraints. In this case, one has 4+2 equations for the 6 cartesian coordinates ($x_i$, $y_i$, $z_i$) and ($x_j$, $y_j$, $z_j$) of the candidate contact points. Redunt constraints must be avoided so that all the equations are linearly independent.

The vector of geometric constraints $\mathbf{\Phi}^G$ is built with $C^2$ continuous functions and the dot and cross product operators do not introduce any discontinuity for an arbitrary point in the domain, therefore, $\mathbf{\Phi}^G$ is a twice-differentiable vector function. The vector equation that contains non-redundant geometric constraints,

$$\mathbf{\Phi}^G(\mathbf{q}^G) = \mathbf{0}, \tag{7}$$

can then be solved using the Newton-Raphson iterative procedure which, theoretically, converges quadratically near the solution. Since $\mathbf{\Phi}^G$ has, at least, a $C^2$ mathematical expression, one can exploit the numerical behaviour with analytical Jacobians.

The distance vector magnitude, $d$, is calculated as the signed Euclidean distance of vector $\mathbf{d}_{\mathrm{PQ}}$, thus, 'negative distances' are considered. At a given time instant, the signed magnitude $d$ indicates one of the three possible contact situtations (Table 1): (i) no contact; (ii) contact at a single point or external contact ($\mathbf{r}_{\mathrm{OP}} = \mathbf{r}_{\mathrm{OQ}}$) and (iii) contact with pseudo-penetration.

| Contact type | Minimum distance $d = sign\left(\mathbf{n}_{i\mathrm{P}} \cdot \mathbf{d}_{\mathrm{PQ}}\right)\left\Vert\mathbf{d}_{\mathrm{PQ}}\right\Vert_2$ |
|---|---|
| No contact | $d > 0$ |
| External contact | $d = 0$ |
| Contact with pseudo-penetration | $d < 0$ |

Table 1: Contact detection situations according to the minimum distance value.

### 3.1 Vector of geometric constraints and analytical Jacobian matrix

There is more than one way to formulate distance computation via geometric contraint equations. The formulation deeply depends on the surface representation. Thus, mathematical modeling for contact detection exploits certain geometric properties associated to the surface representation in order to define the geometric entities (e.g., vectores and surface functions) that participate in the constraint equations. All in all, these equations serve to restrict the solution space to the desired solution, i.e., the solution with most physical meaning for contact.

For 3-D convex objects represented implicitly (Fig. 1), the abovementioned geometric constraints of orthogonality and isosurface are grouped in a 6x1 vector $\mathbf{\Phi}^G$ which is written in the matrix form as,

$$\mathbf{\Phi}^G\left(\mathbf{q}^G\right) = \mathbf{0} \;\Leftrightarrow\; \begin{bmatrix} \mathbf{n}_{OP}^T \cdot \mathbf{t}_{OQ} \\ \mathbf{n}_{OP}^T \cdot \mathbf{b}_{OQ} \\ \mathbf{d}_{PQ}^T \cdot \mathbf{t}_{OQ} \\ \mathbf{d}_{PQ}^T \cdot \mathbf{b}_{OQ} \\ \mathcal{F}_i \\ \mathcal{F}_j \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \tag{7}$$

where

$$\mathbf{q}^G = \left[ x_i, y_i, z_i, x_j, y_j, z_j \right]^T . \tag{8}$$

The last two geometric constraints in Eq. 7 are the implicit surface equations given in the canonical form. i.e., the surface principal axis are aligned with the local coordinate axis and the surface centroid is coincident with the local origin. All vectors in Eq. 7 are defined in global coordinates except for the vector of unknown cartesian coordinates, $\mathbf{q}^G$. In fact, all surface vectors in Eq. 7 depend on $\mathbf{q}^G$ and on the position and orientation of the reference systems. This being, the surface vectors can be expressed in local surface coordinates as (see Fig. 1):

$$\begin{aligned}
\mathbf{d}_{\mathrm{PQ}} &= \mathbf{r}_{\mathrm{OQ}} - \mathbf{r}_{\mathrm{OP}} = \\
&= \mathbf{r}_{\mathrm{O}\beta} + \mathbf{A}_{\mathrm{O}\beta}\mathbf{r}_{\beta j} + \mathbf{A}_{\mathrm{O}\beta}\mathbf{A}_{\beta j}\mathbf{s}_{j\mathrm{Q}} - \left( \mathbf{r}_{\mathrm{O}\alpha} + \mathbf{A}_{\mathrm{O}\alpha}\mathbf{r}_{\alpha i} + \mathbf{A}_{\mathrm{O}\alpha}\mathbf{A}_{\alpha i}\mathbf{s}_{i\mathrm{P}} \right), \\
\mathbf{n}_{\mathrm{OP}} &= \mathbf{A}_{\mathrm{O}\alpha}\mathbf{A}_{\alpha i}\mathbf{n}_{i\mathrm{P}}, \\
\mathbf{t}_{\mathrm{OQ}} &= \mathbf{A}_{\mathrm{O}\beta}\mathbf{A}_{\beta j}\left( \mathbf{n}_{j\mathrm{Q}} \times \mathbf{v}_{j\mathrm{Q}} \right) = \mathbf{A}_{\mathrm{O}\beta}\mathbf{A}_{\beta j}\tilde{\mathbf{n}}_{j\mathrm{Q}}\mathbf{v}_{j\mathrm{Q}}, \\
\mathbf{b}_{\mathrm{OQ}} &= \mathbf{A}_{\mathrm{O}\beta}\mathbf{A}_{\beta j}\left( \mathbf{n}_{j\mathrm{Q}} \times \left( \mathbf{n}_{j\mathrm{Q}} \times \mathbf{v}_{j\mathrm{Q}} \right) \right) = \mathbf{A}_{\mathrm{O}\beta}\mathbf{A}_{\beta j}\tilde{\mathbf{n}}_{j\mathrm{Q}}\tilde{\mathbf{n}}_{j\mathrm{Q}}\mathbf{v}_{j\mathrm{Q}},
\end{aligned} \tag{9}$$

where $\mathbf{n}_{i\mathrm{P}}$, $\mathbf{n}_{j\mathrm{Q}}$, and $\mathbf{v}_{j\mathrm{Q}}$ depend the position vectors of points P and Q in local surface coordinates, $\mathbf{s}_{i\mathrm{P}}$ and $\mathbf{s}_{j\mathrm{Q}}$; the tilde sign (~) above the vectors is the skew-symmetric matrix associated with the corresponding vector; and $\mathbf{v}_{j\mathrm{Q}}$ is an auxiliary vector applied to the calculation of the tangent and binormal vectors to the surface, that is calculated in such a way that is not collinear to the normal vector. In fact, any vector that respects the following two conditions,

$$\mathbf{v} = \mathbf{n} \Rightarrow \mathbf{v} = \mathbf{0} \wedge \mathbf{n} = \mathbf{0}, \tag{10}$$

$$\left| \frac{\mathbf{v}}{\|\mathbf{v}\|} \cdot \frac{\mathbf{n}}{\|\mathbf{n}\|} \right| \neq 1, \tag{11}$$

is a valid auxiliary vector. The first condition (Eq. 10) implies that both vectors are equal if and only if they are the zero vector. As for the second condition (Eq. 11), it implies that the vectors must not be collinear.

The Newton-Raphson method relies on the Taylor series expansion of $\mathbf{\Phi}^G$. This demands the calculation of the geometric contraints Jacobians. Each row of the Jacobian matrix is the first partial derivate of $\mathbf{\Phi}^G$ in order to $\mathbf{q}^G$. Hence,

$$\mathbf{\Phi}^G_{\mathbf{q}^G}\left(\mathbf{q}^G\right) = \frac{\partial}{\partial \mathbf{q}^G}\mathbf{\Phi}^G\left(\mathbf{q}^G\right) \;\Leftrightarrow\; \mathbf{\Phi}^G_{\mathbf{q}^G}\left(\mathbf{q}^G\right) = \begin{bmatrix}
\mathbf{t}^T_{\mathrm{OQ}}\left(\mathbf{n}_{\mathrm{OP}}\right)_{\mathbf{q}^G} + \mathbf{n}^T_{\mathrm{OP}}\left(\mathbf{t}_{\mathrm{OQ}}\right)_{\mathbf{q}^G} \\
\mathbf{b}^T_{\mathrm{OQ}}\left(\mathbf{n}_{\mathrm{OP}}\right)_{\mathbf{q}^G} + \mathbf{n}^T_{\mathrm{OP}}\left(\mathbf{b}_{\mathrm{OQ}}\right)_{\mathbf{q}^G} \\
\mathbf{t}^T_{\mathrm{OQ}}\left(\mathbf{d}_{\mathrm{PQ}}\right)_{\mathbf{q}^G} + \mathbf{d}^T_{\mathrm{PQ}}\left(\mathbf{t}_{\mathrm{OQ}}\right)_{\mathbf{q}^G} \\
\mathbf{b}^T_{\mathrm{OQ}}\left(\mathbf{d}_{\mathrm{PQ}}\right)_{\mathbf{q}^G} + \mathbf{d}^T_{\mathrm{PQ}}\left(\mathbf{b}_{\mathrm{OQ}}\right)_{\mathbf{q}^G} \\
\mathbf{n}^T_{i\mathrm{P}} \qquad \mathbf{0}_{1\times 3} \\
\mathbf{0}_{1\times 3} \qquad \mathbf{n}^T_{j\mathrm{Q}}
\end{bmatrix} \tag{12}$$

where

$$
\begin{aligned}
\left(\mathbf{d}_{\mathrm{PQ}}\right)_{\mathbf{q}^G} &= \mathbf{A}_{\mathrm{O}\beta}\mathbf{A}_{\beta j}\left(\mathbf{s}_{j\mathrm{Q}}\right)_{\mathbf{q}^G} - \mathbf{A}_{\mathrm{O}\alpha}\mathbf{A}_{\alpha i}\left(\mathbf{s}_{i\mathrm{P}}\right)_{\mathbf{q}^G},\\
\left(\mathbf{n}_{\mathrm{OP}}\right)_{\mathbf{q}^G} &= \mathbf{A}_{\mathrm{O}\alpha}\mathbf{A}_{\alpha i}\left(\mathbf{n}_{i\mathrm{P}}\right)_{\mathbf{q}^G},\\
\left(\mathbf{t}_{\mathrm{OQ}}\right)_{\mathbf{q}^G} &= \mathbf{A}_{\mathrm{O}\beta}\mathbf{A}_{\beta j}\left(\mathbf{n}_{j\mathrm{Q}}\times\mathbf{v}_{j\mathrm{Q}}\right)_{\mathbf{q}^G} =\\
&= \mathbf{A}_{\mathrm{O}\beta}\mathbf{A}_{\beta j}\left[\tilde{\mathbf{n}}_{j\mathrm{Q}}\left(\mathbf{v}_{j\mathrm{Q}}\right)_{\mathbf{q}^G} + \tilde{\mathbf{v}}_{j\mathrm{Q}}\left(\mathbf{n}_{j\mathrm{Q}}\right)_{\mathbf{q}^G}\right],\\
\left(\mathbf{b}_{\mathrm{OQ}}\right)_{\mathbf{q}^G} &= \mathbf{A}_{\mathrm{O}\beta}\mathbf{A}_{\beta j}\left(\mathbf{n}_{j\mathrm{Q}}\times\left(\mathbf{n}_{j\mathrm{Q}}\times\mathbf{v}_{j\mathrm{Q}}\right)\right)_{\mathbf{q}^G} =\\
&= \mathbf{A}_{\mathrm{O}\beta}\mathbf{A}_{\beta j}\Big[\tilde{\mathbf{n}}_{j\mathrm{Q}}\tilde{\mathbf{n}}_{j\mathrm{Q}}\left(\mathbf{v}_{j\mathrm{Q}}\right)_{\mathbf{q}^G} - \tilde{\mathbf{n}}_{j\mathrm{Q}}\tilde{\mathbf{v}}_{j\mathrm{Q}}\left(\mathbf{n}_{j\mathrm{Q}}\right)_{\mathbf{q}^G} -\\
&\qquad - \left(\tilde{\mathbf{n}}_{j\mathrm{Q}}\tilde{\mathbf{v}}_{j\mathrm{Q}} - \tilde{\mathbf{v}}_{j\mathrm{Q}}\tilde{\mathbf{n}}_{j\mathrm{Q}}\right)\left(\mathbf{n}_{j\mathrm{Q}}\right)_{\mathbf{q}^G}\Big],
\end{aligned}
\tag{13}
$$

are 3x6 matrices and

$$
\left(\mathbf{s}_{i\mathrm{P}}\right)_{\mathbf{q}^G} = \left[\begin{array}{ccc:ccc} 1 & 0 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0 & 0 \end{array}\right] = \left[\mathbf{I} \mid \mathbf{0}\right], \qquad \left(\mathbf{s}_{j\mathrm{Q}}\right)_{\mathbf{q}^G} = \left[\begin{array}{ccc:ccc} 0 & 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right] = \left[\mathbf{0} \mid \mathbf{I}\right]. \tag{14}
$$

The equations presented in this section are generic to any implicit surface that is $C^2$ continuous. Once equations Eq. 7 and Eq. 12 are implemented, extending the methodology to other surfaces simply requires the deduction of the analytical expressions for $\mathbf{n}_{i\mathrm{P}}$, $\mathbf{n}_{j\mathrm{Q}}$, and $\mathbf{v}_{j\mathrm{Q}}$ $\left(\mathbf{n}_{i\mathrm{P}}\right)_{\mathbf{q}^G}$, $\left(\mathbf{n}_{j\mathrm{Q}}\right)_{\mathbf{q}^G}$, and $\left(\mathbf{v}_{j\mathrm{Q}}\right)_{\mathbf{q}^G}$ which is usually a straightfoward process.

## 4 RIGID BODY SURFACE REPRESENTATIONS AND 3-D VISUALIZATION

The selection of a surface model to represent and visualize the body geometry is of crucial importance for contact analysis. The most important aspects to take under consideration are the geometric representivity of the surface and the analyticity of the surface functions. Preferably, the surface model must provide a compact representation (i.e., a small set of geometric parameters) that can uniquely define a surface. Quadric and superquadrics surfaces are geometric descriptions that are used to model a large variety of 3-D shapes, presenting great shape fidelity for many natural and manmade objects. In this work, the outer surface or certain regions of a bodies boundary are modeled as quadric or superquadric surfaces.

In this work, we will only take into account the implicit surface representation for the contact detection methodology, whereas parametric surface functions are deduced solely for visualization purposes. The surfaces are defined as a polynomial function in $x$, $y$ and $z$ cartesian coordinates. For this reason, such functions are called algebraic surfaces. Quadrics are second-degree polynomials while superquadrics are polynomials with non-negative real expoents. Some of the surface family members are (super)ellipsoids, and one and two sheet (super)hyperboloids. Associated to each surface function are geometric parameters or coefficients that affect the shape, surface dimensions, and overall curvature in a comprehensible manner.

### 4.1 Quadric Surfaces

The implict definition of a quadric surface in the canonical form, i.e., the spatial configuration in which the surface is centered at the origin and the main axis are aligned to the local coordinate system, is expressed as a real valued scalar function,

$$\mathcal{F}(x,y,z) = a_{11}x^2 + a_{22}y^2 + a_{33}z^2 + a_1 x + a_2 y + a_3 z - 1 = 0, \tag{15}$$

with

$$a_{11}a_1 = a_{22}a_2 = a_{33}a_3 = 0, \tag{16}$$

where $\{a_{11}, a_{22}, a_{33}\}$ are shape coefficients and $\{a_1, a_2, a_3\}$ are unit or zero valued coefficients that define the quadric surface family type (Table 2). Dimension parameters along the $x$, $y$, and $z$ directions are given by the following formulas:

$$a = \frac{1}{\sqrt{|a_{11}|}}, \quad b = \frac{1}{\sqrt{|a_{22}|}}, \quad c = \frac{1}{\sqrt{|a_{33}|}}. \tag{17}$$

| Quadric surface type | Coefficients |
|---|---|
| Ellipsoid | $a_{11} > 0$, $a_{22} > 0$, $a_{33} > 0$ |
| Hyperboloid (1 sheet) | $a_{11} > 0$, $a_{22} > 0$, $a_{33} < 0$ |
| Hyperboloid (2 sheets) | $a_{11} < 0$, $a_{22} < 0$, $a_{33} > 0$ |
| Paraboloid (elliptic) | $a_{11} > 0$, $a_{22} > 0$, $a_{33} = 0$, $a_3 < 0$ |
| Paraboloid (hyperbolic) | $a_{11} > 0$, $a_{22} < 0$, $a_{33} = 0$, $a_3 < 0$ |

Table 2: Quadric family classification according to the coefficient values.

Eq. 15 can also be expressed in a matricial form $\mathcal{F}(x,y,z) = \mathbf{x}^T \mathbf{A}\mathbf{x} = 0$, where

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}, \quad \text{and} \quad \mathbf{A} = \begin{bmatrix} a_{11} & 0 & 0 & \frac{1}{2}a_1 \\ 0 & a_{22} & 0 & \frac{1}{2}a_2 \\ 0 & 0 & a_{33} & \frac{1}{2}a_3 \\ \frac{1}{2}a_1 & \frac{1}{2}a_2 & \frac{1}{2}a_3 & -1 \end{bmatrix}. \tag{18}$$

The separation condition heavily depends on the matricial form expressed in the above equation [10,11].

Due to the radial symmetry of the surface, an angle-center parametrization can deduced by expressing the quadric surface in spherical coordinates as:

$$\mathbf{s}_{quad}(\varphi,\theta; a_{11}, a_{22}, a_{33}, a_1, a_2, a_3) = r_{quad}(\varphi,\theta) \begin{bmatrix} \mathrm{s}\theta\ \mathrm{c}\varphi \\ \mathrm{s}\theta\ \mathrm{s}\varphi \\ \mathrm{c}\varphi \end{bmatrix}, \quad \begin{matrix} \varphi \in [\varphi_o, \varphi_i] \\ \theta \in [\theta_o, \theta_i] \end{matrix} \tag{19}$$

with

$$r_{quad}(\varphi,\theta) = \frac{-r_1 + \sqrt{r_1^2 + 4r_2}}{2r_2}, \qquad \begin{cases} r_1 = a_1 \mathrm{s}\theta \mathrm{c}\varphi + a_2 \mathrm{s}\theta s\varphi + a_3 c\theta \\ r_2 = a_{11}\mathrm{s}^2\theta \mathrm{c}^2\varphi + a_{22}\mathrm{s}^2\theta s^2\varphi + a_{33}c^2\theta \end{cases}, \tag{20}$$

where $\mathrm{c} = \cos(.)$, $\mathrm{s} = \sin(.)$, $r_{quad}$ is the radial coordinate, $\varphi$ and $\theta$ are the azimuth and zenith angular coordinates. If $r_{quad}$ = 1 for all the angular domain, we obtain the unit sphere. Note that Eq. 19 is a parametric equation that is valid for all quadric surface family members listed in Table 2. This is extremely important in terms of computational implementation of the visualization of the contact surfaces since only one expression represents all surface members.

Given the implicit representation of the surface in the canonical form (Eq. 15), the normal vector of a quadric surface is the gradient vector of the scalar function $\mathcal{F}$. For surfaces $i$ and $j$, the normal vectors are given by:

$$\mathbf{n}_{i\mathrm{P}} = \nabla\mathcal{F}_i\left(x_i, y_i, z_i\right) = \begin{bmatrix} 2a_{i,11}x_i + a_{i,1} \\ 2a_{i,22}y_i + a_{i,2} \\ 2a_{i,33}z_i + a_{i,3} \end{bmatrix}, \; \mathbf{n}_{j\mathrm{Q}} = \nabla\mathcal{F}_j\left(x_j, y_j, z_j\right) = \begin{bmatrix} 2a_{j,11}x_i + a_{j,1} \\ 2a_{j,22}y_i + a_{j,2} \\ 2a_{j,33}z_i + a_{j,3} \end{bmatrix}. \tag{21}$$

The associated Jacobians are deduced by differential calculus:

$$\left(\mathbf{n}_{i\mathrm{P}}\right)_{\mathbf{q}^G} = 2\left[\begin{array}{ccc:ccc} a_{i,11} & 0 & 0 & 0 & 0 & 0 \\ 0 & a_{i,22} & 0 & 0 & 0 & 0 \\ 0 & 0 & a_{i,33} & 0 & 0 & 0 \end{array}\right], \; \left(\mathbf{n}_{j\mathrm{Q}}\right)_{\mathbf{q}^G} = 2\left[\begin{array}{ccc:ccc} 0 & 0 & 0 & a_{j,11} & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{j,22} & 0 \\ 0 & 0 & 0 & 0 & 0 & a_{j,33} \end{array}\right]. \tag{22}$$

The auxiliary vector is defined as the signed permutation of the normal vector coordinates,

$$\mathbf{v}_{j\mathrm{Q}} = \begin{bmatrix} -n_{j,z} & n_{j,x} & n_{j,y} \end{bmatrix}^T, \tag{23}$$

where $n_{j,x}$, $n_{j,y}$, and $n_{j,z}$ are the normal vector coordinates in the local surface reference system. Note that, any other signed permutation that respects the conditions in Eq. 10 and Eq. 11 is a valid auxiliary. Attention must be made for the possibility of geometric singularities. For example, Eq. 23 presents a geometric singularity for $\mathbf{n} = \tau[1\ \text{-}1\ 1]^T$, with $\tau \in \mathbb{R}$. The Jacobian matrix of $\mathbf{v}_{j\mathrm{Q}}$ is given by:

$$\left(\mathbf{v}_{j\mathrm{Q}}\right)_{\mathbf{q}^G} = 2\left[\begin{array}{ccc:ccc} 0 & 0 & 0 & 0 & 0 & -a_{j,33} \\ 0 & 0 & 0 & a_{j,11} & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{j,22} & 0 \end{array}\right]. \tag{24}$$

In order to increase the efficiency of the contact detection algorithm, proximity queries are usually considered [9]. They consist on simple tests to averiguate if the surfaces are apart for a given instant of time. When dealing with closed surface quadrics, such as spheres and ellipsoids, the most commonly used proximity query is the bounding sphere test. Such technique allows rapid tests for collision detection but are not quite precise, and are often evaluated to determine if a more detailed testing is required. A bounding sphere, as the name suggests, is a 3-D sphere that contains all the points of the quadric surface and share the same centroid. Basically, the test consists on evaluating the inequality between the sum of the semi-major axis of the quadric surfaces and the distance between their centroids. If the distance is lesser or equal to the sum of the semi-major axis then a more detailed testing must be conducted. An elegant and efficient algorithm for detecting contact between two ellipsoids was presented by Choi et al. [11]. This algorithm is based on the separation condition of two ellipsoids, which is a necessary and sufficient condition that states that the characteristic equation has positive roots if and only if the ellipsoids do not have common interior points [10].

### 4.2 Superquadric Surfaces

Superquadrics are a generalization of quadric surfaces with the expoent replaced by an arbitrary non-negative number. Relatively to quadric surfaces, the expoent consists in the introduction of a new degree of freedom for geometric modeling. By varying the expoent value one can create rounded, squared, filleted or pinched shapes. In the particular case of superellipsoids, since the proposed contact methodology refers to convex objects, the surface shape is mediated between circular and rectangular shapes.

The implicit surface representation of a superquadric, in the canonical form, is given by the following expression:

$$\mathcal{F}(x,y,z) = \left(a_{11}x^{\gamma_2} + a_{22}y^{\gamma_2}\right)^{\frac{\gamma_1}{\gamma_2}} + a_{33}z^{\gamma_1} - 1 = 0, \tag{25}$$

where $\{a_{11}, a_{22}, a_{33}\}$ are shape coefficients, and $\gamma_1$ and $\gamma_2$ are the expoents. Depending on the signal value of shape coefficients, one can define the family type of the superquadric (Table 3). If $\gamma_1 = \gamma_2$ with $\gamma_1, \gamma_2 \in \{1,2\}$ then Eq. 25 falls in to a quadric implicit function. Barr [12] introduced a spherical product operator through which a surface is created given two parametric curves that lay upon two orthogonal planes. In fact, it is merely a generalization of the spherical to rectangular coordinate transformation. Basically, $\gamma_1$ and $\gamma_2$ are the expoents the orthogonal curves that lay on yOz and xOy, respectively. The formulas for the dimension parameters along the x, y, and z directions are given by:

$$a = \left(\frac{1}{|a_{11}|}\right)^{\frac{1}{\gamma_2}}, \quad b = \left(\frac{1}{|a_{22}|}\right)^{\frac{1}{\gamma_2}}, \quad c = \left(\frac{1}{|a_{33}|}\right)^{\frac{1}{\gamma_1}}. \tag{26}$$

| Superquadric surface type | Coefficients |
|---|---|
| Superellipsoid | $a_{11} > 0$, $a_{22} > 0$, $a_{33} > 0$ |
| Superhyperboloid (1 sheet) | $a_{11} > 0$, $a_{22} > 0$, $a_{33} < 0$ |
| Superhyperboloid (2 sheets) | $a_{11} > 0$, $a_{22} < 0$, $a_{33} < 0$ |

Table 3: Superquadric family classification according to the coefficient values.

The angle-center parametrization is deduced in the same way as for the quadric surfaces. This being, the parametric superquadric representation $\mathbf{s}_{super}$ is given by:

$$\mathbf{s}_{\text{super}}(\varphi,\theta;a_{11},a_{22},a_{33},\gamma_1,\gamma_2) = r_{\text{super}}(\varphi,\theta)\begin{bmatrix} \mathrm{s}\theta\ \mathrm{c}\varphi \\ \mathrm{s}\theta\ \mathrm{s}\varphi \\ \mathrm{c}\varphi \end{bmatrix}, \quad \begin{array}{l} \varphi \in [\varphi_o, \varphi_i] \\ \theta \in [\theta_o, \theta_i] \end{array} \tag{27}$$

with

$$r_{\text{super}}(\varphi,\theta) = \frac{1}{\left(\mathrm{s}^{\gamma_1}\theta\left(a_{11}\mathrm{c}^{\gamma_2}\varphi + a_{22}\mathrm{s}^{\gamma_2}\varphi\right) + a_{33}\mathrm{c}^{\gamma_1}\theta\right)^{\frac{1}{\gamma_1}}}, \tag{28}$$

where $r_{super}$ is the radial coordinate, $\varphi$ and $\theta$ are the azimuth and zenith angular coordinates. Note that Eq. 27 is a parametric equation that is valid for all the mentioned superquadric family members in Table 3.

Barr's [12] implicit definition of the superquadric surface is composed by a function raised to $\gamma_1/\gamma_2$. By twice differentiating Eq. 25, the expoent $\gamma_1/\gamma_2 - 2$ appears in the resulting expressions. As a consequence, only values of $\gamma_1 \geq 2\gamma_2$ are permissible, so that non-negative expoentes are preserved. This deeply hampers the desired geometric representativity and excludes the particular case of the quadric surfaces. For contact analysis with implicit surfaces, the superquadric definition proposed by Barr is quite limited, although it is entirely applicable when considering the parametric version of the geometric constraints. Therefore, an alternative quadric surface generalization is considered:

$$\mathcal{F}\left(x, y, z\right) = a_{11}x^{\gamma_1} + a_{22}y^{\gamma_2} + a_{33}z^{\gamma_3} - 1 = 0, \tag{29}$$

where $\{a_{11}, a_{22}, a_{33}\}$ are shape coefficients, and $\gamma_1$, $\gamma_2$, and $\gamma_3$ are real non-negative expoents.

In any computational implementation involving polynomials with rational expoents, one must be aware of the numerical evaluation of superquadric equations [16]. The correct order of evaluation of these exponential terms is $\left(x^2\right)^{1/\varepsilon}$, with $\gamma = 2/\varepsilon$, to assure that the result is not a complex number when $x < 0$. In order to prevent divisions by zero, the expoent must be greater or equal to zero. The values of $\gamma_1$, $\gamma_2$, and $\gamma_3$ are normally bounded between 2 and infinity so that only convex shapes, with no geometric singularities, are modeled.

The normal vector to a superquadric surface is also given by the gradient vector and the Jacobian matrices are obtained in the same fashion:

$$\mathbf{n}_{i\mathrm{P}} = \nabla\mathcal{F}_i\left(x_i, y_i, z_i\right) = \begin{bmatrix} a_{i,11}\gamma_{i,1}x_i\left(x_i^{\,2}\right)^{\frac{1}{\varepsilon_{i,1}}-1} \\ a_{i,22}\gamma_{i,2}y_i\left(y_i^{\,2}\right)^{\frac{1}{\varepsilon_{i,2}}-1} \\ a_{i,33}\gamma_{i,3}z_i\left(z_i^{\,2}\right)^{\frac{1}{\varepsilon_{i,3}}-1} \end{bmatrix}, \quad \mathbf{n}_{j\mathrm{Q}} = \nabla\mathcal{F}_j\left(x_j, y_j, z_j\right) = \begin{bmatrix} a_{j,11}\gamma_{j,1}x_j\left(x_j^{\,2}\right)^{\frac{1}{\varepsilon_{j,1}}-1} \\ a_{j,22}\gamma_{j,2}y_j\left(y_j^{\,2}\right)^{\frac{1}{\varepsilon_{j,2}}-1} \\ a_{j,33}\gamma_{j,3}z_j\left(z_j^{\,2}\right)^{\frac{1}{\varepsilon_{j,3}}-1} \end{bmatrix}, \tag{30}$$

$$\left(\mathbf{n}_{i\mathrm{P}}\right)_{\mathbf{q}^G} = \left[\begin{array}{ccc:ccc} a_{i,11}\gamma_{i,1}\left(\gamma_{i,1}-1\right)\left(x_i^{\,2}\right)^{\frac{1}{\varepsilon_{i,1}}-1} & 0 & 0 & 0 & 0 & 0 \\ 0 & a_{i,22}\gamma_{i,2}\left(\gamma_{i,2}-1\right)\left(y_i^{\,2}\right)^{\frac{1}{\varepsilon_{i,2}}-1} & 0 & 0 & 0 & 0 \\ 0 & 0 & a_{i,33}\gamma_{i,3}\left(\gamma_{i,3}-1\right)\left(z_i^{\,2}\right)^{\frac{1}{\varepsilon_{i,3}}-1} & 0 & 0 & 0 \end{array}\right],$$

$$\left(\mathbf{n}_{j\mathrm{Q}}\right)_{\mathbf{q}^G} = \left[\begin{array}{ccc:ccc} 0 & 0 & 0 & a_{j,11}\gamma_{j,1}\left(\gamma_{j,1}-1\right)\left(x_j^{\,2}\right)^{\frac{1}{\varepsilon_{j,1}}-1} & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{j,22}\gamma_{j,2}\left(\gamma_{j,2}-1\right)\left(y_j^{\,2}\right)^{\frac{1}{\varepsilon_{j,2}}-1} & 0 \\ 0 & 0 & 0 & 0 & 0 & a_{j,33}\gamma_{j,3}\left(\gamma_{j,3}-1\right)\left(z_j^{\,2}\right)^{\frac{1}{\varepsilon_{j,3}}-1} \end{array}\right]. \tag{31}$$

The problem in generalizing the proposed contact methodology to superquadrics is that they lead to polynomial equations with fractional expoents, which are very difficult to solve numerically. When two local surface coordinates are close to zero, the Jacobian matrices in Eq. 31 become ill-conditioned which can potentially jeoperdize the non-singularity of the Jacobian matrix of geometric contraints (Eq. 12). Unfortunatly, an elegant proximity query, such as the separation condition for ellipsoids, is not, to our knowledge, available. Bounding spheres are the only proximity query here considered.

### 4.3 Fitting superquadric surfaces to a set of points

To apply the proposed contact methodology for mechanical systems with freeform surfaces, superquadrics need to be fitted in the neighbourhood of the points of contact. Notice that for complex objects, a decomposition into sub-objects is necessary for effective superquadric approximation [9]. Superquadric fitting consists on determining the geometric coefficients of the surface that minimizes the Euclidean distances between the modeled shape and each point of the data set. After selecting an appropriate superquadric implicit function to model the point data, the problem of approximating the 3-D object reduces to the least square minimization of the non-linear algebraic function over the space of parameters of the function. The approximation procedure relies on the minimization of an error-of-fit function which has the general form:

$$LMS = \sum_{k=1}^{N} \left( \mathcal{F}(x_k, y_k, z_k) - 1 \right)^2, \tag{32}$$

where $N$ is the number of points and $\mathcal{F}$ the superquadric implicit function.

The Levenberg–Marquardt algorithm [13,14] is a robust approximation method, commonly used for surface fitting, that confers a numerical solution to the problem of minimizing a non-linear function. Being an iterative procedure, it is necessary to provide an initial guess for the surface parameters. For highly non-linear error-of-fit functions, the algorithm converges only if the initial guess is close to the final solution.

## 5 CONTACT FORCE MODELS

The computational simulation of the contact phenomenon is strongly dependent on the reactive force model used to represent the interaction between the system components and its environment. Together with the systems dynamical response, the contact force model must consider the material and geometric properties of the colliding bodies, and account for some level of energy dissipation. These characteristics are ensured with a continuous contact force model, in which the deformation and contact forces are considered to be continuous functions during the complete period of contact. The response of the dynamic system is obtained by simply including updated forces into the equations of motion. The contact force models can include elastic, viscous, and friction force components. Friction will not be considered in this work.

The Hunt-Crossley contact force model extended Hertz contact law, which is purely elastic, as a non-linear function of the penetration that includes energy loss due to internal damping, in which the histeresis damping factor expresses the energy dissipated during an impact. According to Hunt and Crossley [5], the contact normal force is given by the sum of the elastic and damping forces:

$$F_N = K\delta^n + \lambda\delta^n\dot{\delta} \tag{33}$$

where $K$ is the generalized stiffness parameter, $\delta$ is the (pseudo-)penetration value that is equal to calculated minimum Euclidean distance $d$, $\dot{\delta}$ is the relative normal penetration velocity, and $\lambda$ the histeresis damping factor. The expoent $n$ is close to unity and depends on the contacting surface geometries. The generalized stiffness depends on the material properties (Poisson ratio and the Young modulus) and on the geometry of the contacting bodies. As shown in Eq. 33, damping also depends on the penetration. This assures that the contact forces present no discontinuity at zero valued penetration. The model proposed by Lankarani and Nikravesh is a particular case of the one proposed by Hunt and Crossley. On the other hand, the coefficient of restituition depends on the energy dissipated during impact and, consequently, also relates to the coefficient of damping [15]. This being, Lankarani and Nikravesh established, for central direct contacts with no friction, the following expressions for coefficient of restituition $e$ and relative initial impact velocity $\dot{\delta}_0$:

$$e = 1 - \frac{2\lambda}{3K}\dot{\delta}_0 \qquad (34)$$

The force expressed by Eq. 33, when drawn versus penetration, results in a hysteresis loop as shown in Fig. 2. The area of the hysteresis loop is equal to the energy loss due to the internal damping of the material. The hysteresis damping function assumes that the loss in energy during impact is due to the material damping of t he colliding bodies, which dissipates energy in the form of heat.

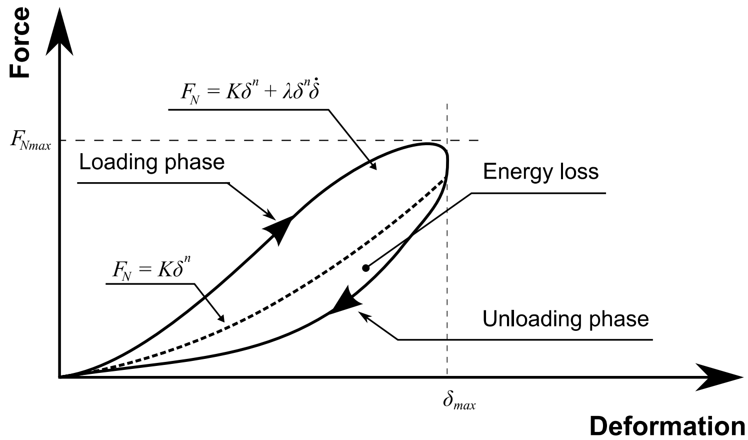


Figure 2: Hunt and Crossley contact force model.

The occurrence of surface overlap is the basis to evaluate the local deformation of the contacting bodies. Then, a constitutive contact law, such as the continuous contact force model proposed by Hunt and Crossley or by Lankarani and Nikravesh can be applied to evaluate the contact force developed in the direction perpendicular to the plane of collision. Within the contact region, the surface deformation and resulting contact stresses are approximated by a function of the proximity (i.e., the overlap) of the originally undeformed surfaces.

## 6 CONTACT DETECTION ALGORITHM

Within the framework of multibody dynamics, the contact detection algorithm takes the bodies positions and orientations calculated from the equations of motion, and returns the location of the pair of contact points together with the contact forces. The algorithm consists on the following steps:

(i) Establish the time interval $[t_0, t_{end}]$ for the analysis;
(ii) At time step $t_0$, establish the initial conditions for the position vector, $\mathbf{q}_0 = \mathbf{q}(t_0)$, velocity vector, $\dot{\mathbf{q}}_0 = \dot{\mathbf{q}}(t_0)$, and contact candidate pair of points $\mathbf{q}_0^G = \mathbf{q}^G(t_0)$;
(iii) Evaluate the proximity queries; if the surfaces are sufficiently close then go to (iv), otherwise go to (vii);
(iv) Run the Newton-Raphson method with analytical Jacobians to obtain the vector of the contact points $\mathbf{q}^G$;
(v) Compute the signed distance magnitude $d$ and check for contact; if there is contact, evaluate contact forces; if not go to step (vii);
(vi) Add the contact forces to the vector of applied forces $\mathbf{g}$;
(vii) Solve the equations of motion deduced from the multibody dynamics formulation in order to obtain the body positions and orientations for the new time step $t + \Delta t$;
(viii) Update the system time variable $t$ and use the vector $\mathbf{q}^G$ obtained in (iii) as the initial guess for the Newton-Raphson method;
(ix) Go to step (iii) and proceed with the whole process for the new time step;
(x) Exit the main algorithm's loop when the final time step is reached.

Note that the contact detection algorithm is ran $n$ times for each time step, where $n$ is the number of rigid contact pairs within the multibody system. Fig. 3 shows the flowchart of the contact methodology. The numerical implementation of this methodology leads to an efficient algorithm since the information of the previous time step is used as an initial guess to find the solution of the non-linear equations and, therefore, only a few iterations are required to obtain the solution.

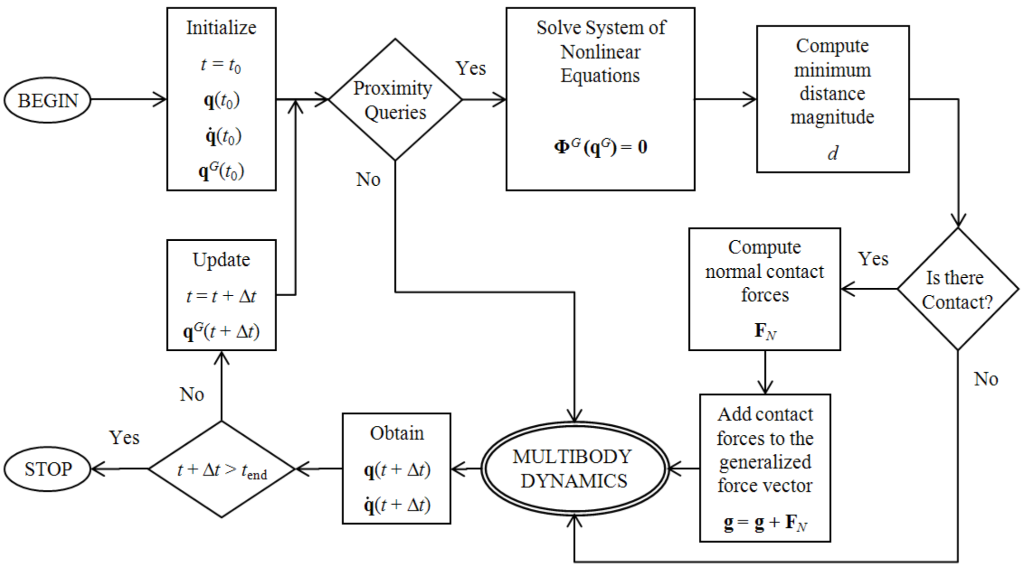


Figure 3: Flowchart of the proposed contact detection algortithm.

## 7 RESULTS

The results here presented refer to the distance computation between a contact pair represented as ellipsoidal (quadric) surfaces as depicted in Fig. 4(A). No proximity queries or multibody dynamics calculations are undertaken in order to evaluate the distance computation efficiency during the analysis. The contact analysis is performed for a time interval of 200 timesteps with a tolerance of $10^{-6}$ for the Newton-Raphson accuracy. Both surfaces are initially aligned with the global coordinate system axes, and the centroids of surface $i$ and $j$ are positioned at the origin and at $[0\ 4\ 0]^T$, respectively. The prescribed motion induced upon surface $i$ and $j$ are a 360º rotation along the X axis and along the vector $[1\ 1\ 1]^T$, respectively, as represented in Fig. 4(B). The initial approximation, $\mathbf{q}^G(t_0)$, consisted on the half distance between the centroids. For the remaning timesteps, the Newton-Raphson approximations are the resulting vector from the previous timestep, $\mathbf{q}^G(t)$. The analysis evaluation took 0.9960 seconds, corresponding to a total of 987 Newton-Raphson iterations required for the 200 timesteps, without diverging. Approximately 5 iterations per timestep are necessary to solve the non-linear system of equations (Eq. 7). Fig. 5(A) and Fig. 5(B) show the influence of the initial approximation of the Newton-Raphson method in which the initial points for $t$ = 0 did not lie on the surfaces. Both plots show that the residual error follows, approximately, a quadratic convergence. When the approximation is closer to the solution, a supra-quadratic convergence is noticed. A test was performed for distance computation between a contact pair of superquadrics which failed when the coordinates were close to zero. However, for a contact pair made up by a quadric and a superquadric the method did not fail and presented a computational behaviour close to the quadric case here presented.

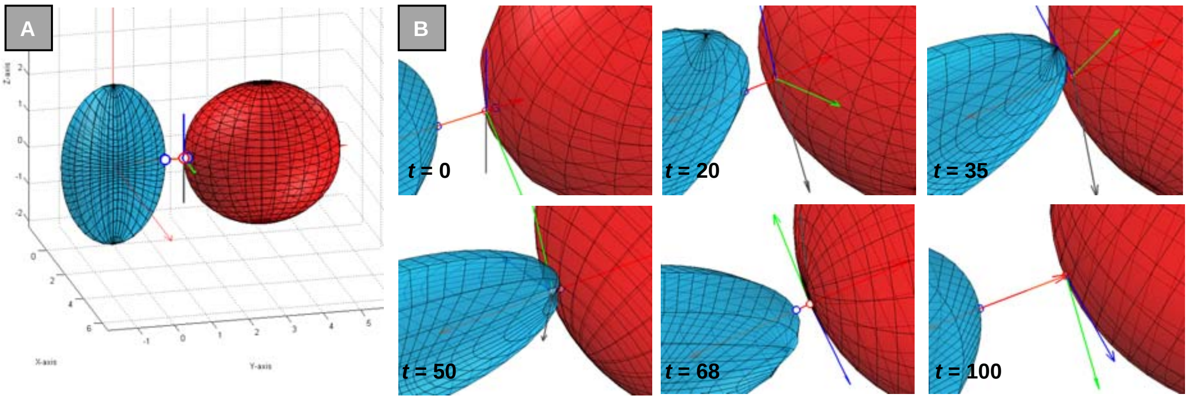


Figure 4: (A) Initial configuration of the surface pair. (B) 3D graphics of the contact region and its vinicity in six different timesteps. (blue surface – $i$, red surface – $j$; red vectors – normals, grey vector – auxiliary, blue vector – tangent, green vector – binormal)

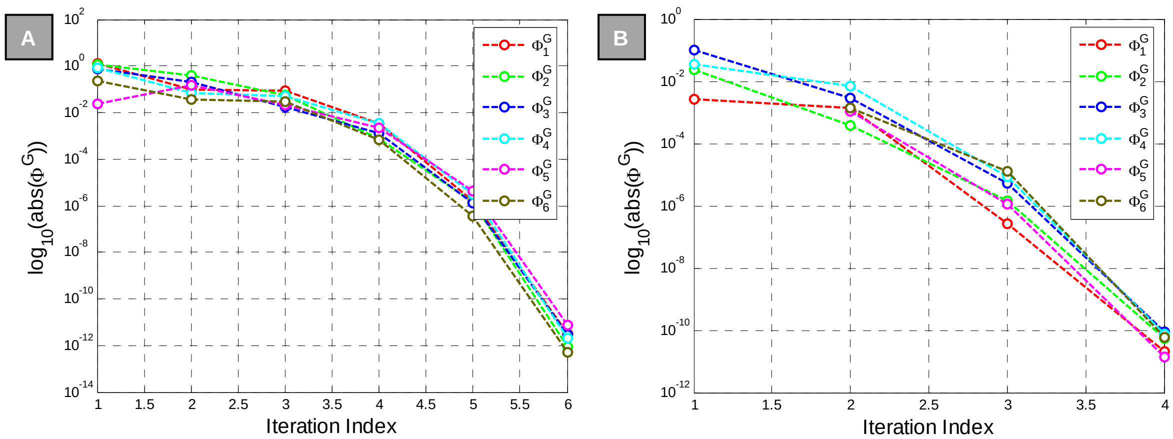


Figure 5: (A) Semi-log plot of vector $\mathbf{\Phi}^G$ at $t$ = 0. (B) Semi-log plot of vector $\mathbf{\Phi}^G$ at $t$ = 50.

## 8 DISCUSSION

The proposed contact methodology relies on a simple geometric principle and on locus contraints that are quite intuitive. Consequently, the methodology is easily formulated according to vector calculus and algebraic and differential geometry.

From a mathematical point of view, quadrics detain a better behaviour than superquadrics since special treatment is not required for continuity singularities and the issues associated with rational expoents do not take place (divisions by zero and the appearance of complex numbers for negative domain values). On the other hand, superquadrics possess a higher geometric representativity since the varying expoents control the overall curvature of the surface, contrary to quadrics that have a constant power. Notice that the contact formultion proposed here is limited to convex surfaces. In fact, if one or both of the surfaces are concave, multiple solutions can be obtained for the geometric constraint vector. In order to apply the contact detection metholodoly to concave surfaces, a set of superquadrics can be arranjed to fit the non-convex surface. Quadrics and superquadrics have a small number of geometric parameters and approximate a wide class of convex objects.

The auxiliary vector chosen for quadric surfaces provides constant Jacobian matrices and a geometric singularity. For superquadrics it is necessary to guarantee that the normal vector and auxiliary vector Jacobians are not ill-conditioned, i.e., with not too many zeros, so that $\mathbf{\Phi}_{\mathbf{q}^G}^G$ does not become a singular matrix when a pair of local coordinates are close to zero. As an alternative, instead of considering the implicit surface definition, the parametric representation provides normal vector, tangent and, binormal information available continuosly over the surface, even across edges that are very square.

Contact calculations contribute quite significantly to the computational cost of multibody dynamics analysis. The evaluation of functions describing the superquadrics involves the computation of trigonometric and exponential functions, and is thus expensive. Proximity queries help attenuate the computational burden but for superquadric surfaces an efficient test, such as the separation condition for ellipsoids, is not yet available. The usage of analytical Jacobians, besides guaranteeing the geometric accuracy of the result, also contributes to the computational efficiency, since no matrix estimate is required for each iteration.

All in all, the computational efficieny and robustness are the major advantages of the present model is that it converges rapidly, allowing simulations to be performed interactively. Though the computational time is dependent on the initial guess (a limitation inherited from the Newton-Raphson algorithm) 200 timesteps were completed in less than 1.5 seconds.

## 9 CONCLUSIONS

In this work, an accurate, efficient, and easily implementable algorithm for minimum distance computation between quadric and superquadric surfaces is presented. This contact algorithm does not resort on optimization methods or convex polyhedra geometries, making use of analytical expressions for the surface vectors and associated Jacobian matrices. A mathematical artifice was introduced to compute tangent and binormal vectors for implicit surfaces given the normal vector to a surface. The methodology provides a uniform framework for distance computation between objects described as arbitrary convex implicit surfaces but geometric singularities still persist. The speed at which distance computation is performed enables real-time dynamic simulations.

The proposed contact methodology explores the implicit representation of superquadrics which relies on algebraic expressions with rational expoents, contrary to the parametric counterpart which involves trigonometric functions raised to rational expoents, i.e., expressions with higher non-linearity. The limitations of the conventional superquadric implicit function are addressed, leading to further instigation for alternative surface models, and to the consideration of the parametric representation for contact analysis with superquadrics. Despite the aforementioned limitations, quadric and superquadric surfaces are relatively easy to handle mathematically.

Instead of presenting specific parametric expressions for each (super)quadric surface member [12], the generic (all quadric members and superellipsoid and superhyperboloid members) angle-center formulas of the parametrized surfaces given by Eq. 19 and Eq. 17 were deduced and are bestowed for visualization and 3-D modeling purposes.

The terms ‘locus contraint’ and ‘isosurface constraint’ are introduced in contact analysis.

As future works, we aim to extend this approach to conformal (or concave) objects modeled as unions of convex shapes; to develop a non-collinear vector with no geometric singularities; and to exploit the contact detection methodology for surfaces with a parametric representation.

## ACKNOWLEDGMENTS

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